Chain-of-Thought Steering and Land-Interaction Testing for Tropical-Cyclone Forecasting

TrackFormer v10–v34: from a CNN Steering-Field Encoder to a Frozen-Backbone Causal Isolation of Land Drag

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Abstract

We forecast the full state of a tropical cyclone — storm motion, maximum wind, central pressure, radius of maximum wind, and the twelve 34/50/64-kt wind-radii — at twenty six-hourly lead times (6–120 h) using an explicit atmospheric steering-flow representation. Starting from a small CNN encoder that reads a deep-layer-mean steering-wind patch around the storm (v10–v20), we introduce a *chain-of-thought* (CoT) decomposition that predicts the steering flow itself as an explicit, inspectable intermediate representation before deriving track from it (v21, v22), then condition that estimate on its own recent history (t-24 h, t-12 h, now) to let recurvature respond to how the flow is *evolving* rather than only its instantaneous value. This last model, **v23**, reaches **434.96 km** RMS track error on a WP+EP 2020+ held-out set restricted to full 20-lead-horizon windows (3,763 windows, 72% land-covered on the WP-only subset) — the best result in this line of twenty-five versions. We then test two further physically-motivated hypotheses the same way. Four attempts to add more environmental signal on top of v23 (an environmental token, an ocean-heat CNN patch, a drift adapter, and raw ERA5 steering wind fed directly) all come back null: once a CoT representation already extracts the steering signal that matters, handing the model the raw field again is redundant. Three architecturally different attempts at a land/terrain-interaction correction (uniform training, oversampled training, storm-normalized oversampling) converge on an aggregate null once we isolate the correction from ordinary retrain noise by *freezing* v23's own trained weights and training only a ∼437-parameter gated addition on top — though it still splits 1–1 against v23 on the two real out-of-training storms available. This frozen-backbone isolation technique is the paper's methodological contribution: it makes visible that retrain-to-retrain seed noise (∼19 km/seed) is large enough to manufacture or hide most of the effects reported in this line of work, and that a large in-distribution test set does not always agree with genuinely out-of-training real-storm validation. An appendix summarizes the earlier, foundational protected dual-stream architecture and random-matrix uncertainty head (v1–v9) that this line of work builds on.

Roadmap. Section 1 is the paper's primary content: the CNN steering encoder, chain-of-thought steering, the temporal steering stack (v23), four null environmental additions, and the three-attempt land-interaction test resolved by a frozen-backbone isolation (v31–v34), evaluated throughout on the WP+EP 2020+ full-horizon test set. Appendix A summarizes the earlier protected dual-stream Transformer and random-matrix uncertainty head (v1–v9) that established the codebase and the all-basin/WP 2020+ dataset this line inherited.

1 Chain-of-thought steering and land-interaction testing (v10–v34)

This section evaluates twenty-five versions on WP+EP storms from 2020 onward, restricted to windows with the full 20-lead horizon (3,763 windows, 72% land-covered on the WP-only subset) — a larger and more demanding held-out set than Appendix A's all-basin/WP split (Table 2), built to test that appendix's own closing prediction: that the next lever past a history-only floor would be a compact environmental-steering branch fused strongly to track.

A CNN steering-field encoder (v10–v20). A small convolutional encoder (conv, GELU, strided downsampling) reads a 17×17 patch of deep-layer-mean steering wind (a weighted blend of 850/500/200-hPa u, v) centered on the storm — the same family of field-encoder design as Appendix A's ERA5 branch, now applied to a physically motivated *derived* steering field rather than raw multi-level reanalysis. Successive versions repaired data alignment bugs, added a four-term track loss, EMA and stronger regularization, and an intensity-persistence baseline with rarity/speed reweighting (v13–v19), converging on v20's deep-layer-mean steering patch.

Chain-of-thought steering (v21, v22). Rather than regressing track displacement directly from an opaque decoder, v21 first predicts the ambient *steering flow itself* as an explicit intermediate representation, then derives the track from it — a chain-of-thought (CoT) decomposition that makes the underlying physical mechanism (storms move with the steering flow, plus a learned recurvature correction) an explicit step in the computation graph rather than an implicit function of a black-box decoder. v22 extends this to a *latent* CoT with two weight-tied feedback rounds that iteratively refine the same steering representation before the track head reads it, in the spirit of universal-transformer-style recurrence.

Temporal steering stack — v23, the best model in this line. Formally, where $\boldsymbol{f}_\tau$ is the CNN steering-field encoding at time τ, v21's chain-of-thought head first derives an explicit steering estimate before deriving track from it,

$$\hat{\boldsymbol{u}}_{\text{steer}} = g_{\text{CoT}}(\boldsymbol{f}_{t_0}), \qquad \widehat{\Delta \boldsymbol{p}}_\ell = h_\ell\big(\hat{\boldsymbol{u}}_{\text{steer}},\ \boldsymbol{H}^{\text{kin}}\big), \tag{1}$$

so the steering estimate is a named, inspectable quantity rather than an implicit intermediate activation. v23 conditions this on a short *history* of the same representation, $t-24\,\text{h}$, $t-12\,\text{h}$, and now, rather than the instantaneous value alone:

$$\hat{\boldsymbol{u}}_{\text{steer}} = g_{\text{CoT}}\big(\boldsymbol{f}_{t_0},\ \boldsymbol{f}_{t_0-12\text{h}},\ \boldsymbol{f}_{t_0-24\text{h}}\big), \tag{2}$$

so recurvature can be conditioned on how the steering flow has been *evolving*, not only its instantaneous value — a discrete analogue of estimating the flow's local time-derivative alongside its value. This is the best-performing model across the entire v10–v34 line: 434.96 km RMS track error (10-seed ensemble) on the WP+EP 2020+ full-horizon test set, a further ∼30% reduction over the already-improved Appendix-A numbers in Table 2 (on this section's harder, larger test set).

Environmental additions on top of v23 — v24, v25, v27, v28, v29: null or marginal. With the CoT steering representation already in place, four further attempts to add environmental signal all failed to improve on v23: an environmental token (ocean heat content, deep-layer shear, mid-level humidity; v25), an ocean-heat CNN patch (GODAS) feeding the intensity head (v27), an

N-only meridional drift adapter (v28, whose matched-retrain control v28abl confirmed the apparent gain was ordinary run-to-run noise), and — most tellingly — feeding *raw* 0.25° ERA5 steering wind (u, v at 200/550/850 hPa) directly on top of v23 (v29): 438.70 km, +3.74 km worse, every one of five seeds above the v23 bar. Once a CoT representation already extracts the steering signal that matters from a given class of field, handing the model the raw field a second time is largely redundant: *the right representation of a signal matters more than an additional raw copy of that signal.*

Land interaction: three from-scratch attempts, then a frozen-backbone isolation (v31–v34). Motivated by real-world reports of typhoons stalling at mountainous coastlines (Typhoon Gaemi, 2024, held up roughly a day at Taiwan's Central Mountain Range) and terrain-deflection mixture-of-experts literature (AOT-TCNet, arXiv 2603.29200), three architecturally different attempts add a small along-track land/terrain correction on top of v23, of the general form

$$\Delta \boldsymbol{p}_\ell \mathrel{+}= g(\boldsymbol{x}^{\text{land}}) \cdot a_{\max} \tanh\big(\boldsymbol{k}(\boldsymbol{x}^{\text{land}}) \cdot \boldsymbol{W}_i\big) \cdot \hat{\boldsymbol{v}}_0, \qquad \boldsymbol{x}^{\text{land}} = (d_{\text{near}}, d_{\text{ahead}}, h_{\text{ahead}}, |\boldsymbol{v}_0|, |\text{lat}|), \quad (3)$$

where $\boldsymbol{k}(\cdot)$ is a small MLP producing four knot values interpolated to all 20 leads by a fixed basis $\boldsymbol{W}_i$, $\hat{\boldsymbol{v}}_0$ is the current heading unit vector, and $g(\cdot) \equiv 1$ for v31–v33 (the correction is unconditionally on) or a separate learned, zero-biased sigmoid gate for v34, described below. Both the expert $\boldsymbol{k}$ and, where present, the gate g are zero-initialized so every version starts as exactly v23:

- **v31** (uniform training): a `LandDrag` module — distance-to-land and terrain-height-ahead through a zero-initialized MLP producing an along-track push — trained end-to-end from scratch. Result: 443.07 km, +8.11 km worse than v23, every seed above the bar.
- **v32** (window-level oversampled near-land training, larger push cap): backfired badly, 460.48 km (+25.52), worst *exactly* in the mountainous-near-land bucket the fix targeted — consistent with oversampling a small number of storms' repeated windows rather than a general effect.
- **v33** (storm-normalized oversampling, closing v32's pseudo-replication problem): closed most of the gap, 442.33 km (+7.37, the closest of the three from-scratch attempts), and — on the two real out-of-training storms available for a genuinely independent check, Typhoon Tip (1979) and Typhoon Noul (2026) — *beat* v23 on both.
- **v34** (mixture-of-experts gate on a *frozen*, already-trained v23 backbone): every prior attempt retrained the whole architecture from scratch each time, confounding any real land effect with ordinary retrain-to-retrain noise (quantified just below at $\sim$19 km/seed). v34 instead loads a real, already-trained v23 checkpoint and freezes every parameter except a new $\sim$437-parameter `LandGate`: the same along-track expert as `LandDrag`, plus a learned sigmoid gate deciding how much the expert contributes per sample (both zero-initialized, so the model starts as exactly v23). This is a true same-backbone-plus-one-addition comparison. Result: -0.33 km against its own frozen v23 backbone — essentially null — and a bucketed breakdown by terrain height and along/cross-track error confirms the null holds even in the mountainous-near-land regime every earlier attempt targeted (+1.0/-6.2 km at 48 h/120 h). On the same two real storms: v34 *beats* both v23 and v33 on Tip 1979 (795 vs 939/876 km) but is worse than both on Noul 2026.

Table 1: Land-interaction line: aggregate WP+EP 2020+ full-horizon test (3,763 windows) and the two independent, genuinely out-of-training real-storm checks. Aggregate deltas are against v23 (v31–v33) or against v34's own frozen v23_seed0 backbone (v34, the causally clean comparison). Best per column in **bold**; real-storm columns show whichever model wins that storm.

model	aggregate (km)	Tip '79 (km)	Noul '26, ocean/landfall (km)
v23 (baseline)	**434.96**	939	267 / 356
v31 (LandDrag, uniform)	443.07 (+8.11)	—	—
v32 (LandDrag, window-oversampled)	460.48 (+25.52)	—	—
v33 (LandDrag, storm-normalized)	442.33 (+7.37)	**876**	**243** / **319**
v34 (LandGate, frozen backbone)	460.52 (−0.33 vs. backbone)	**795**	287 / 382

Methodological lessons.

1. *Seed noise dominates small version deltas.* A single retrained v23 seed scored 460.85 km on the same test set — 25.89 km worse than the 10-seed ensemble mean, from ordinary retrain variance alone, comparable in size to most of the "effects" reported above. Any claimed improvement under ∼20 km from a from-scratch retrain warrants a matched-seed-count or frozen-backbone check before being taken as real.

2. *Freezing a real backbone isolates causal effects that comparing two from-scratch runs cannot.* v34's frozen-backbone design is what let this line of work state a definitive null on the land-drag hypothesis after three inconclusive from-scratch attempts; the technique generalizes to any small architectural addition under test.

3. *The in-distribution aggregate test set does not always agree with real-storm validation.* v33 and v34 are both null-to-slightly-negative on the large WP+EP aggregate, yet each beats v23 on one of the two individually-checked, genuinely out-of-training real storms. Small-n real-storm checks cannot replace an aggregate metric, but they surface disagreements the aggregate alone would miss.

Reproducibility. Dataset builders, training scripts, checkpoints, and evaluation are at `https://github.com/yu314-coder/typhoon-predict`. This line's models are trained on Google Colab (T4/A100); Appendix A's earlier models trained on Apple GPU (Metal/MPS).

A Foundational architecture: protected dual-stream TrackFormer (v1–v9)

This appendix summarizes the earlier line of work this paper builds on: the dataset, model, and uncertainty head that established the codebase and its all-basin/WP 2020+ evaluation split (Table 2), before Section 1's steering-flow and land-interaction testing became the project's main line of inquiry.

Dataset. IBTrACS v04r01 (all basins, 1980+), 84,150 windows: a history of H=9 steps at 6-h spacing and a target of L=20 future 17-dimensional states (per-step E/N motion, vmax, pressure, RMW, twelve 34/50/64-kt wind radii) at leads $\{6, \ldots, 120\}$ h. Split by storm to prevent leakage (train $\leq$ 2015, validation 2016–2019, test $\geq$ 2020); inputs standardized on training-split statistics only, with explicit presence-flag channels for missing measurements.

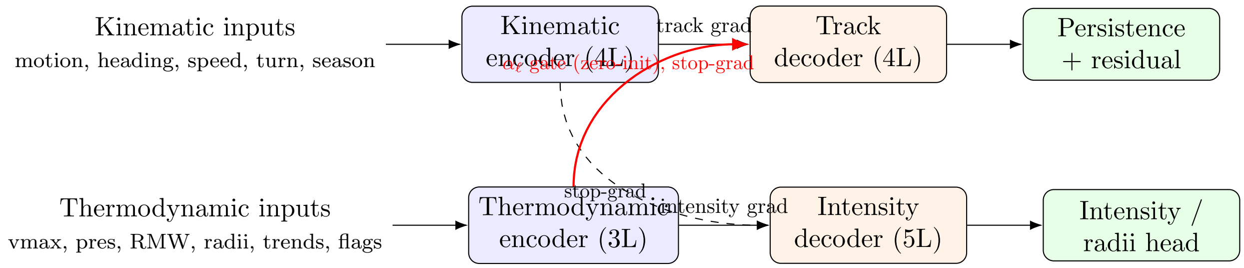

Figure 1: TrackFormer's protected dual-stream design. Solid arrows carry task gradients back only into their own encoder; thermodynamics reaches track *only* through a zero-initialized, per-lead gated adapter on detached features, so at initialization the model is exactly a protected track-only predictor.

Negative transfer. A single-stream Transformer trained on 40 base features reaches the "old" row of Table 2. Adding 8 motion-dynamics features (heading, speed, turn-rate, intensity trends; "v2") improves intensity/pressure/RMW/radius substantially but *raises* track error (720.0 $\rightarrow$ 737.3 km), and up-weighting the track loss makes it worse still (806.6 km) — the signature of gradient conflict over shared capacity, not a loss-weighting problem. A paired storm-level bootstrap confirms the regression is statistically tied ($\widehat{\Delta}_{\text{track}} = +17.7\,\text{km}$, 95% CI $[-22.2, +60.5]\,\text{km}$) while the intensity gains are large and consistent: the shared representation, not the added features, is the problem.

Protected dual-stream architecture. Kinematic and thermodynamic features are encoded separately (Figure 1); the track decoder cross-attends only to the kinematic encoding, and thermodynamics reaches the track head only through a zero-initialized gated adapter,

$$\boldsymbol{H}_\ell^{\text{track}} \;=\; \text{TrackDec}(\boldsymbol{q}_\ell, \boldsymbol{H}^{\text{kin}}) \;+\; \alpha_\ell\, \mathcal{A}\big(\text{sg}(\bar{\boldsymbol{H}}^{\text{th}})\big), \qquad \alpha_\ell \overset{\text{init}}{=} 0, \tag{4}$$

where $\text{sg}(\cdot)$ is stop-gradient. So intensity can help track (if it lowers track error) but can never harm it. Track itself is a damped constant-velocity persistence baseline plus a zero-initialized learned recurvature residual, so the network begins as pure persistence and learns only the deviation from it. Track loss is a lead-weighted Huber on per-step motion plus a cumulative Huber, never down-weighted by predicted variance; intensity uses masked Huber + Gaussian NLL with a monotone wind-radii penalty.

Random-matrix block-structured uncertainty head. A full 340×340 output covariance is statistically ill-posed and would re-couple track to thermodynamics, so we impose a block-diagonal $\boldsymbol{\Sigma} = \text{blkdiag}(\boldsymbol{\Sigma}_{\text{track}}, \boldsymbol{\Sigma}_{\text{int}}, \boldsymbol{\Sigma}_{\text{rad}})$ with a matrix-normal track block and low-rank+diagonal thermodynamic blocks. Sample eigenvalues are cleaned with a two-point generalized sample-covariance model: the companion Stieltjes transform solves a cubic

$$az\,m^3 + \big[a(z-y+1)+z\big]m^2 + \big[a+z-y+1-y\beta(a-1)\big]m + 1 = 0, \tag{5}$$

whose loss of a real root marks the noise bulk's support edge λ_+; eigenvalues above λ_+ are retained as signal, the rest replaced by a positive floor (setting $a{=}1$ recovers the classical Marchenko–Pastur edge $\lambda_+ = (1+\sqrt{y})^2$). Fit on independent storm-block residuals and applied only after the mean heads converge, this sharpens calibration without touching the point forecast.

Table 2: Held-out test performance (lower is better). Track is RMS plane distance over 6–120 h; the rest are masked MAE. All models evaluated on identical target windows. Best per column in **bold**.

split	model	track (km)	vmax (kt)	pres (hPa)	RMW (km)	radius (km)
WP 2020+	single-stream, 40-feat	720.0	22.08	21.24	12.85	31.54
	single-stream, 48-feat (v2)	737.3	**21.49**	**17.68**	**11.77**	30.94
	TrackFormer (v3)	**659.0**	21.61	18.06	11.81	**28.83**
	TrackFormer (v8, +data)	**648.8**	**20.69**	**15.94**	**11.28**	**27.83**
	TrackFormer (v9, +env)	**618.4**	**18.55**	15.76	11.52	**27.15**
all-basin 2020+	single-stream, 40-feat	638.6	20.32	18.01	15.28	29.99
	single-stream, 48-feat (v2)	644.2	19.55	15.38	**14.18**	29.31
	TrackFormer (v3)	592.1	18.59	14.61	14.87	27.85
	TrackFormer (v8, +data)	580.5	17.90	**13.34**	**13.89**	**27.10**
	TrackFormer (v9, +env)	**543.4**	**16.73**	14.29	14.15	27.80

Results summary. The protected architecture (v3) lowers WP-2020+ track from 720.0 to 659.0 km (−8.5%, storm-bootstrap Pr(better) = 0.995) while *improving* intensity/pressure/radius. Keeping partial-lead windows (storm-end/short-storm fixes, masked rather than discarded) doubled clean training windows and produced v8, best on every metric (648.8 km WP). Adding a third *environmental* stream of IBTrACS-derived absolute position, distance-to-land, and a climatological SST proxy — since the dual-stream model is translation-invariant and cannot otherwise use latitude (Coriolis, recurvature, SST) — produced v9, the best model in this appendix's line (618.4 km WP, 543.4 km all-basin; Figure 2). The consistent lesson: **data diversity** > **engineered features** > **parameters** (a 17.7M single-stream model overfit and did worse than a 3.3M one; a track-only model matched an ERA5-fed one). This is what motivated Section 1's environmental-steering branch, this paper's main subject.

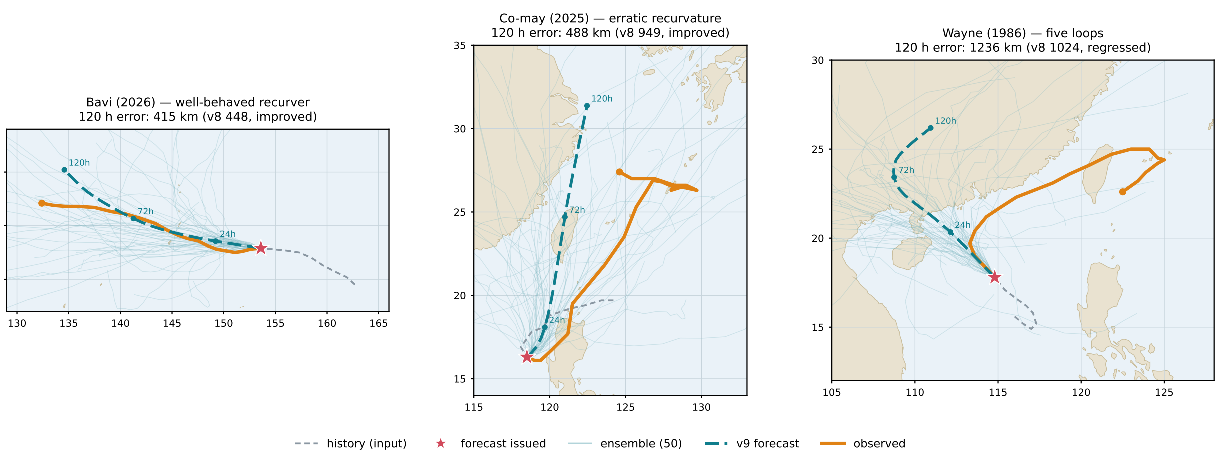

Figure 2: TrackFormer v9 (triple-stream) on three case studies, each a 120-hour forecast (teal, dashed) issued from an early fix (star) against the observed track (orange), with the 50-member RMT ensemble (faint). Absolute latitude sharpens the well-behaved *Bavi* (2026) and *halves* the error on the erratic *Co-may* (2025), but cannot tame *Wayne*'s (1986) five loops, where climatology is no substitute for the real-time steering flow. Bavi and Co-may are out-of-sample ($\geq$2020); Wayne (1986) is in the training window and shown as a stress test.